

UE23CS320B - Capstone Project Phase - 2

SEMESTER - VI

END SEMESTER ASSESSMENT

Project Title : Integrating Polygenic Risk Score and Digital Twin for Coronary Artery Disease Prediction

Project ID : PW26_SVM_06

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Abstract and Scope

1. Project Introduction

This project introduces a hybrid Digital Twin framework for personalized Coronary Artery Disease (CAD) risk prediction. By integrating population-calibrated Polygenic Risk Scores (PRS) with multi-modal clinical and lifestyle data, the system moves beyond static risk assessment. It leverages a machine learning-based counterfactual engine to not only predict individual CAD risk but to dynamically simulate the impact of targeted behavioral and clinical interventions.

2. Problem Statement

CAD is India's leading cause of mortality, with South Asians developing the disease a decade earlier and facing up to four times the risk of Western populations. Despite this burden, existing clinical tools are primarily validated on European cohorts and ignore the unique genetic architecture of the Indian demographic. Furthermore, current predictive models are static and monolithic. They fail to account for inherited genetic predispositions and offer no counterfactual capabilities to simulate how specific lifestyle or medical changes might alter a patient's risk trajectory, leading to systematic underdiagnosis and missed intervention opportunities.

Abstract and Scope

3. Overview of Scope

The scope encompasses the end-to-end development of a multi-modal, explainable CAD prediction pipeline tailored for the Indian risk profile. Key deliverables include:

- **Genomic Calibration (PRS):** Implementing a Polygenic Risk Score calibrated specifically to Indian allele frequencies via Monte Carlo simulations to accurately baseline inherited cardiovascular risk.
- **Multi-Modal Integration:** Harmonizing a foundational lifestyle dataset of 70,000 records with 1,190 clinical instances to create a comprehensive patient profile.
- **ML Counterfactual Engine (Digital Twin):** Developing a predictive framework that models the patient's current state and enables dynamic "what-if" simulations. This allows clinicians to alter input variables mathematically (e.g., stabilizing blood pressure or ceasing smoking) to instantly recalculate and project revised risk probabilities.
- **Explainable AI (XAI):** Integrating transparency mechanisms to isolate and explicitly communicate which specific genetic, lifestyle, or clinical factors are driving both the baseline prediction and the counterfactual outcomes.

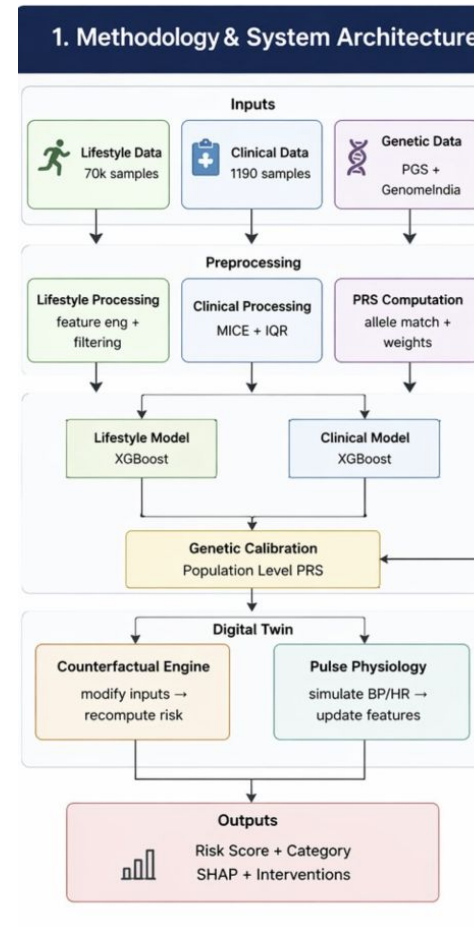
Suggestions from Phase 2- review 2

1. Implement a dynamic update architecture to process longitudinal, real-world patient data (e.g., follow-up clinical visits or continuous wearable inputs) and dynamically update the risk baseline and intervention projections.
2. Evaluate the integration of BioGears and look into existing Digital Twin models which can be implemented.

Implementation Details

Methodology

- Multi-stage pipeline integrating lifestyle (70K), clinical (1,190), and genetic data
- Independent model training for lifestyle and clinical cohorts
- Population-level Polygenic Risk Score (PRS) computed using SNP alignment and allele frequencies
- Machine learning models evaluated; XGBoost selected based on best AUC performance
- PRS incorporated as a calibration offset rather than a feature
- Final prediction computed as model probability (85%) plus genetic baseline (15%)
- SHAP-based explainability used for feature-level and domain-level attribution
- Digital Twin module implemented using counterfactual intervention analysis



System Architecture

- Multi-layer architecture integrating lifestyle, clinical, and genetic data
- Input layer includes lifestyle, clinical, and genetic data streams
- Preprocessing layer applies independent cleaning and feature engineering pipelines
- Genomic module performs SNP matching and Polygenic Risk Score (PRS) computation
- Model layer consists of parallel lifestyle and clinical prediction models
- Calibration layer applies PRS-based adjustment along with Platt scaling
- Output layer generates risk probability and risk stratification
- Digital Twin layer enables what-if simulations for lifestyle interventions

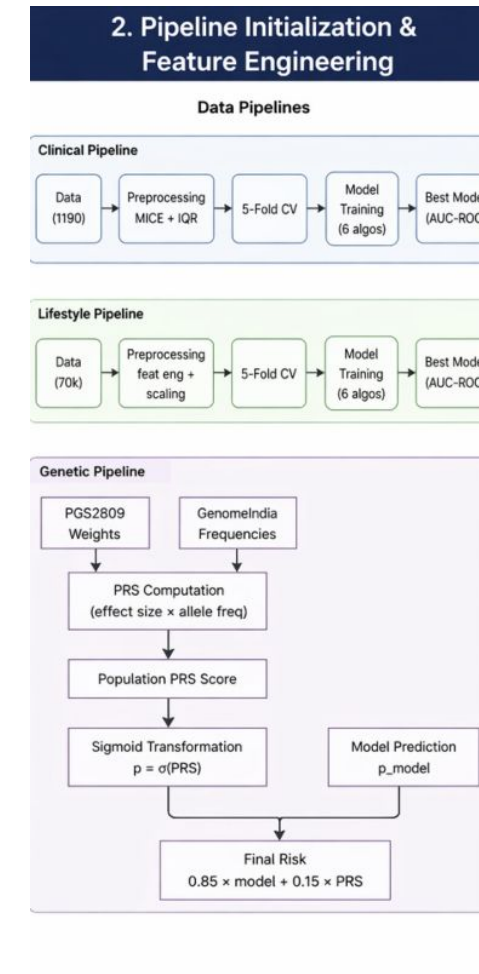
Implementation Details: Pipeline Initialization & Feature Engineering

Pipeline Initialization

- Structured as a modular, multi-stage pipeline
- Processes lifestyle, clinical, and genetic data independently
- Core stages include preprocessing, PRS computation, model training, and risk integration
- Final layers add explainability and digital twin-based intervention analysis

Feature Engineering

- Lifestyle features: BMI derived from height & weight, cleaned blood pressure, and encoded behavioral factors (smoking, alcohol, activity)
- Clinical features: merged multi-source dataset with imputation (MICE) and outlier handling (IQR); includes continuous medical variables
- Genomic features: SNP alignment (182/205 retained) used to compute a population-level PRS



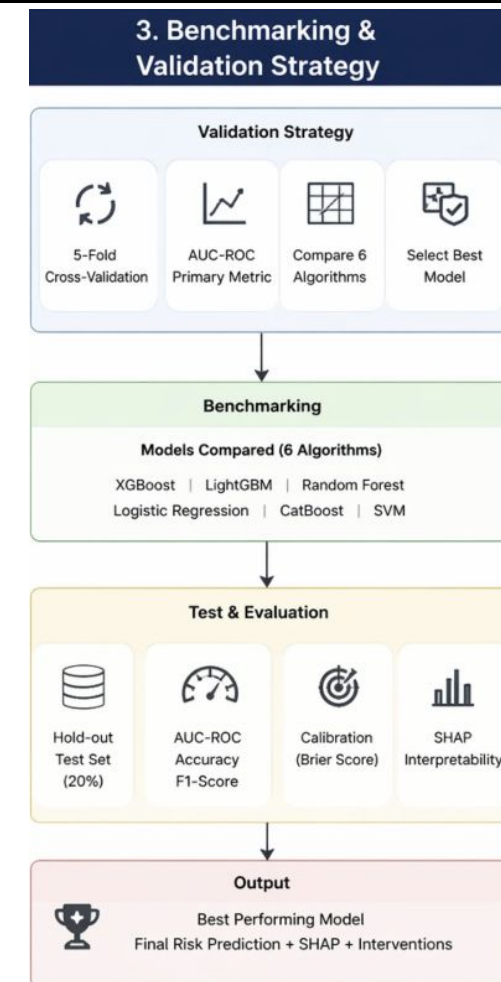
Implementation Details: Benchmarking & Validation Strategy

Benchmarking

- Compared our model with standard risk scores like Framingham
- Focused on two key things:
 - How well the model separates high vs low risk (AUC)
 - How accurate the predicted probabilities are (calibration)
- Evaluated results separately for lifestyle and clinical models
- Clinical model showed stronger prediction performance

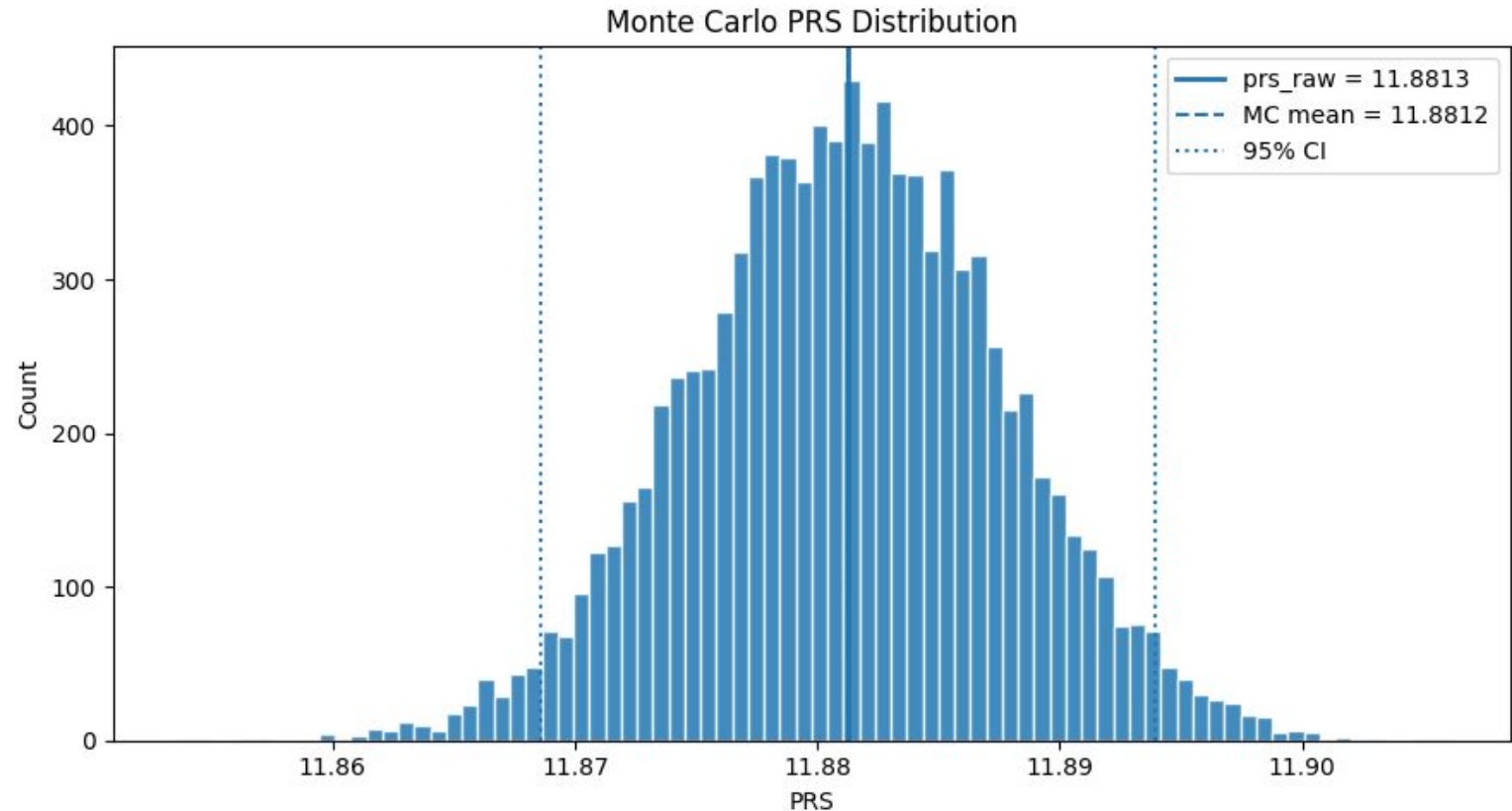
Validation

- Used cross-validation to ensure stable model performance
- Final evaluation done on a separate test set
- Applied probability calibration for reliable predictions
- **Digital Twin:**
 - a. Verified risk changes behave as expected
 - b. Ensured outputs are consistent and reasonable

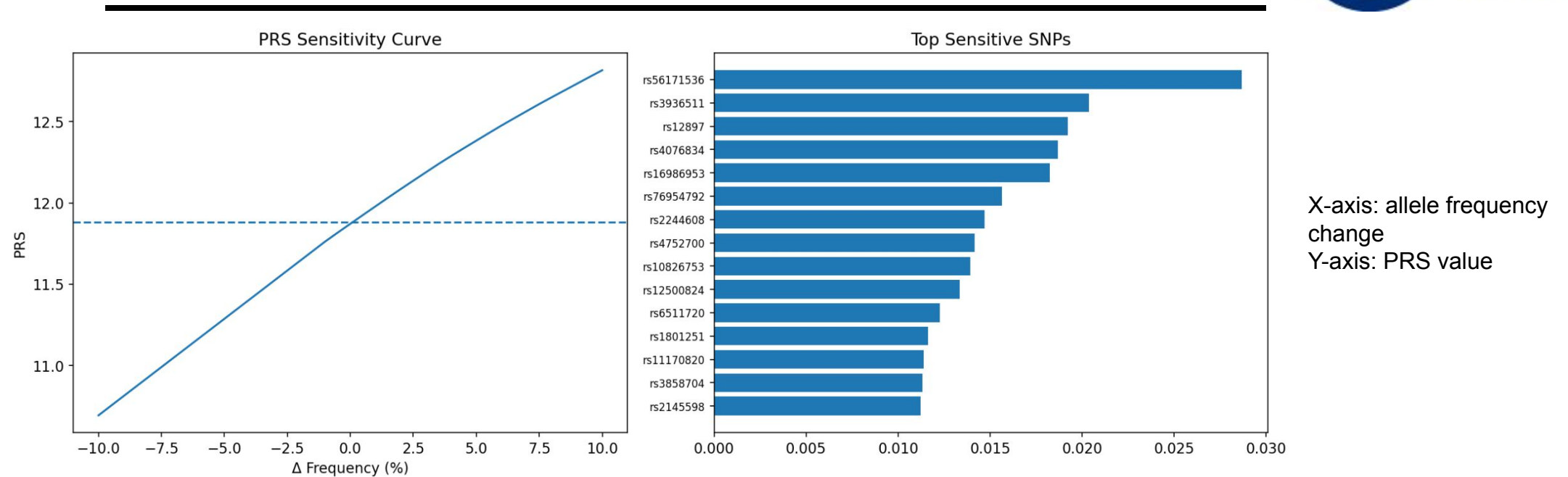


Results : PRS calculation

- PRS distribution generated using Monte Carlo simulation (10K individuals)
- Mean (μ) and confidence interval computed
- Individual PRS positioned relative to population baseline



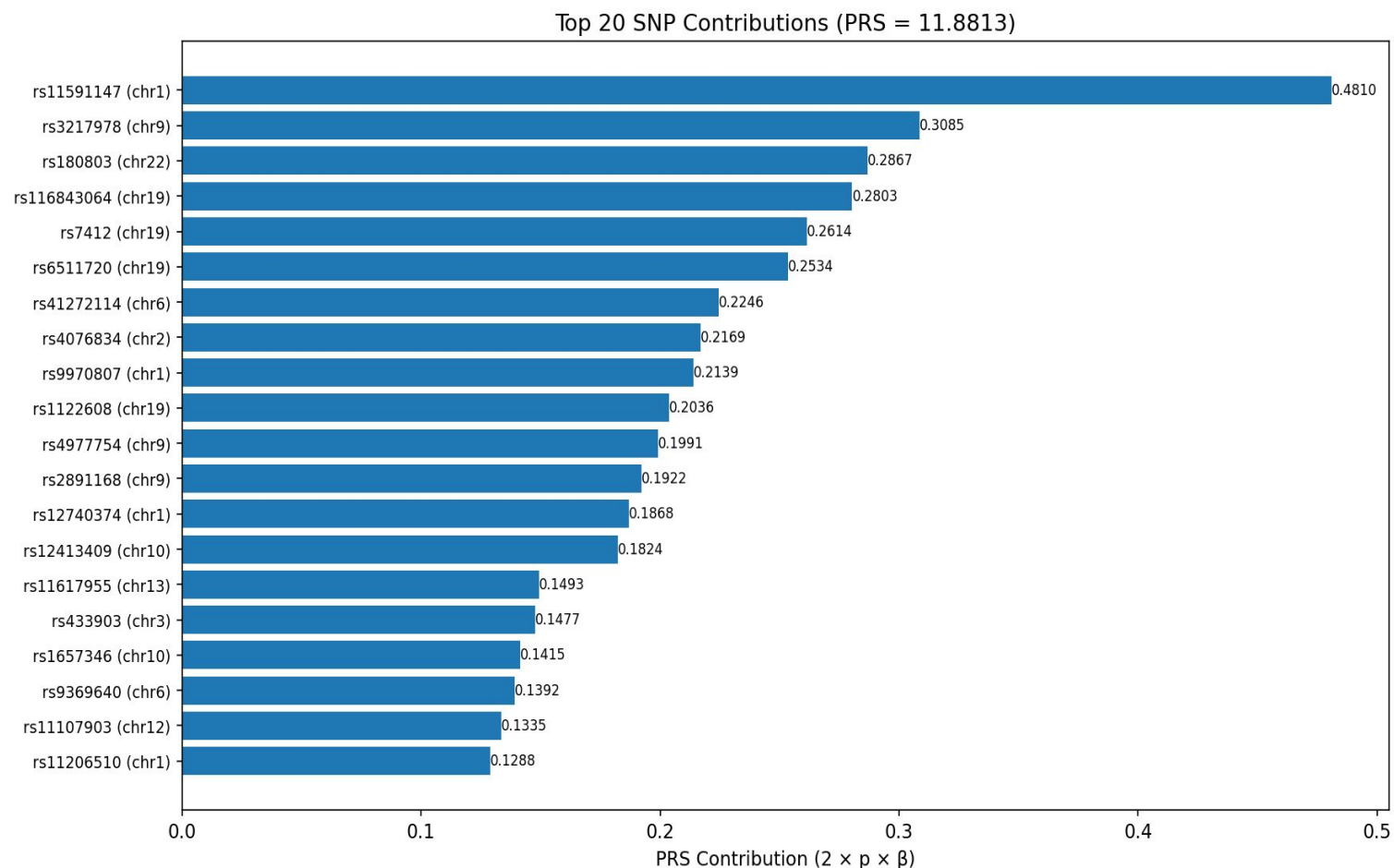
Results : PRS calculation



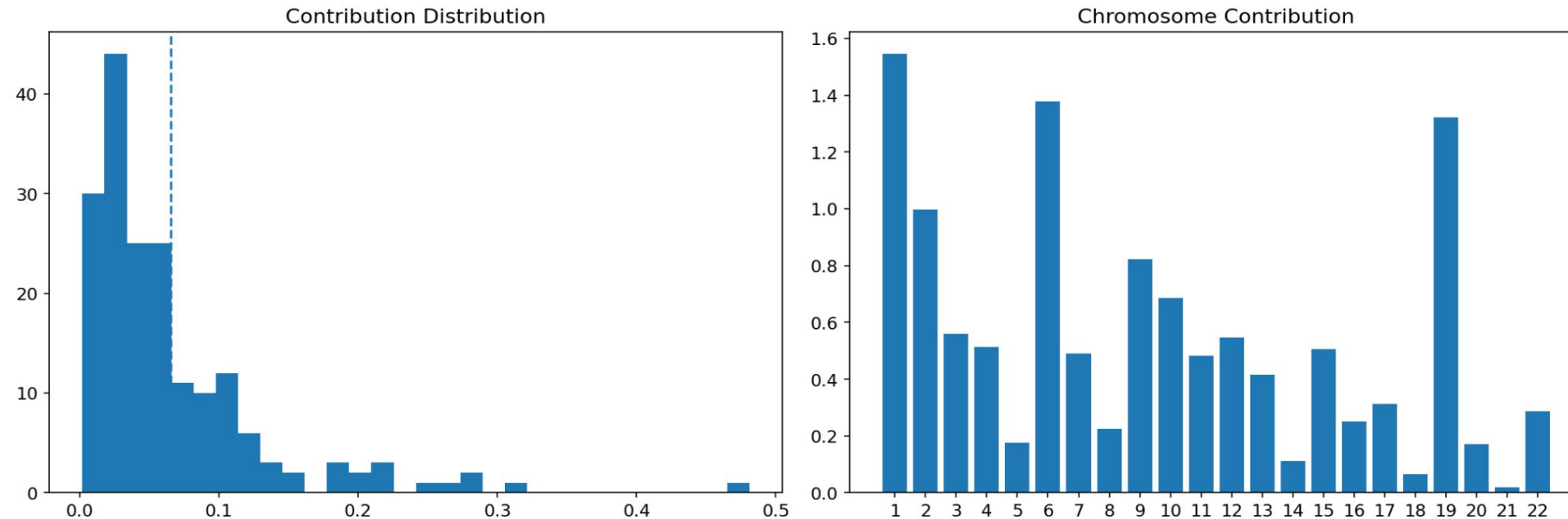
- PRS increases linearly with allele frequency changes
- No abrupt jumps meaning stable and robust model
- Dashed line is baseline PRS
- Shows SNPs with highest impact on PRS when perturbed
- Few SNPs drive major sensitivity changes
- Majority have moderate influence

Results : PRS calculation

- Displays the top 20 SNPs contributing to the Polygenic Risk Score (PRS)
- Contribution calculated using:
$$PRS\ contribution = 2 \times p \times \beta$$
- SNPs are ranked by magnitude of contribution
- Each bar shows how much that SNP adds to total genetic risk



Results : PRS calculation



- Most SNPs have very low contribution, with a few showing higher impact
- Distribution is right-skewed (long-tail), consistent with typical polygenic behavior
- Mean contribution is indicated by the dashed reference line
- Contribution varies across chromosomes and is not uniformly distributed
- Certain chromosomes (such as 1, 6, and 19) show higher influence
- Suggests localized genetic regions driving risk

Results : Lifestyle model

- XGBoost achieved the highest AUC (approximately 0.80)
- All models show low variance, indicating stable performance
- XGBoost selected as the final model based on overall performance

Model Comparison (Cross-Validation)		
Model	AUC	F1
Logistic Regression	0.79	0.71
Random Forest	0.79	0.71
XGBoost	0.80	0.72
SGD	0.77	0.69

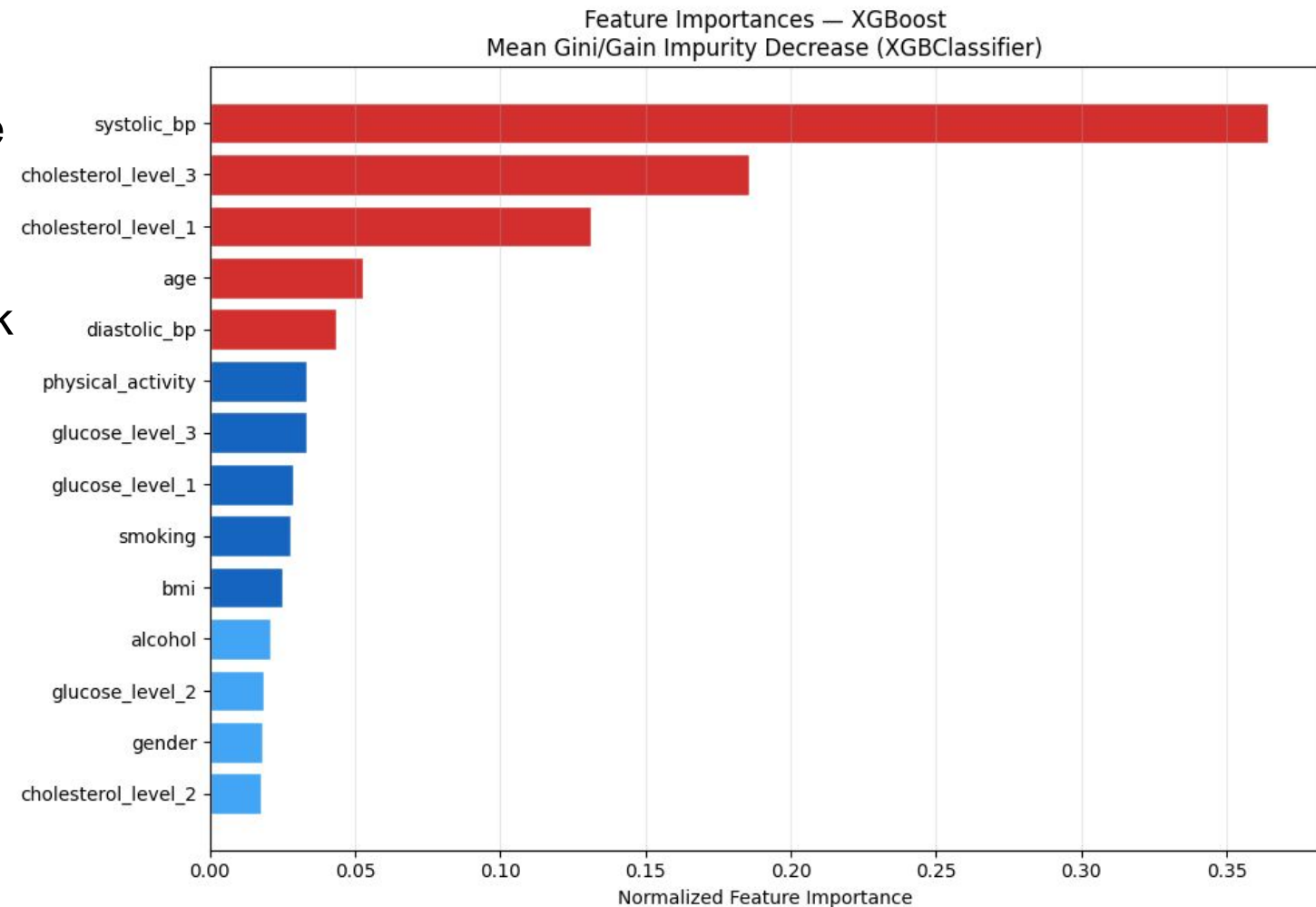
Results : Lifestyle model

- XGBoost achieves the highest AUC (0.806), indicating best discrimination
- Lowest Brier score (0.178), showing best calibration among all models
- Other models perform competitively but are slightly less accurate and less well-calibrated
- XGBoost selected as the final model based on overall performance

Final Model Evaluation (Test Set)		
Model	AUC	Brier Score
Logistic Regression	0.797	0.185
Random Forest	0.798	0.183
XGBoost	0.806	0.178
SGD	0.792	0.188

Results : Lifestyle model

- Systolic BP is the most influential feature (~36%), followed by cholesterol levels
- Top features align with known clinical risk factors (BP, cholesterol, age)
- Lifestyle factors (smoking, BMI, activity) show moderate but meaningful impact
- Model captures both clinical and behavioral contributors to CAD risk



Results: Clinical Model Comparison (Cross-Validation)

- XGBoost achieves the highest AUC (approximately 0.88), indicating best discrimination
- Gradient Boosting shows similar performance but with slightly higher risk of overfitting
- All models demonstrate stable cross-validation performance
- XGBoost selected as the final model based on overall performance

Clinical Model Comparison (Cross-Validation)		
Model	AUC	F1
Logistic Regression	0.85	0.79
Random Forest	0.87	0.79
XGBoost	0.88	0.81
Gradient Boosting	0.88	0.81

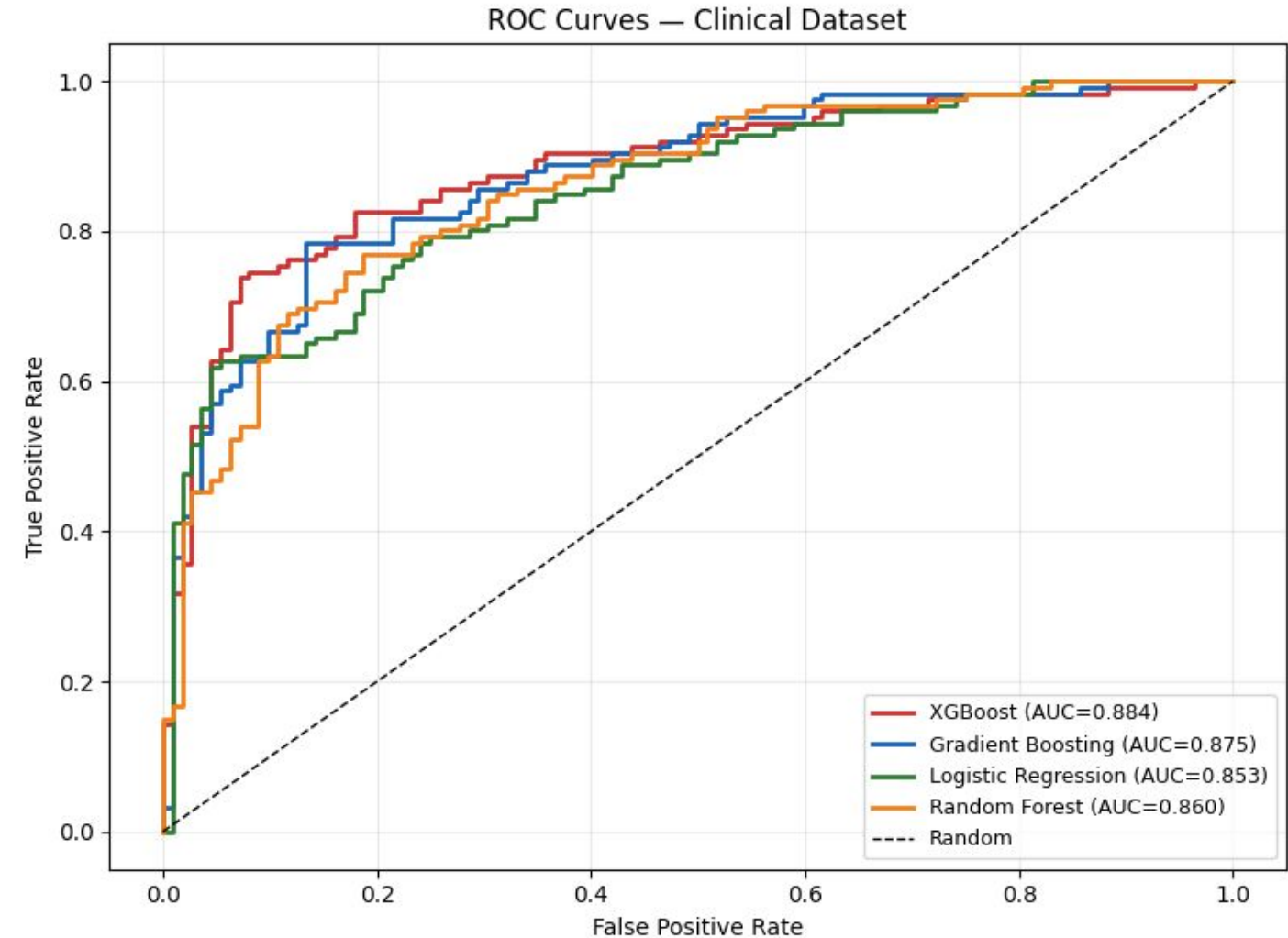
Results : Final Clinical Model Performance

- XGBoost achieves the highest AUC (0.88), indicating strong discrimination
- Lowest Brier score (0.13), showing best calibration among models
- Gradient Boosting performs similarly but with slightly higher Brier score
- XGBoost selected as the final clinical model based on overall performance

Final Model Performance (Clinical)		
Model	AUC	Brier
XGBoost	0.88	0.13
Gradient Boosting	0.88	0.14
Logistic Regression	0.85	0.16
Random Forest	0.86	0.15

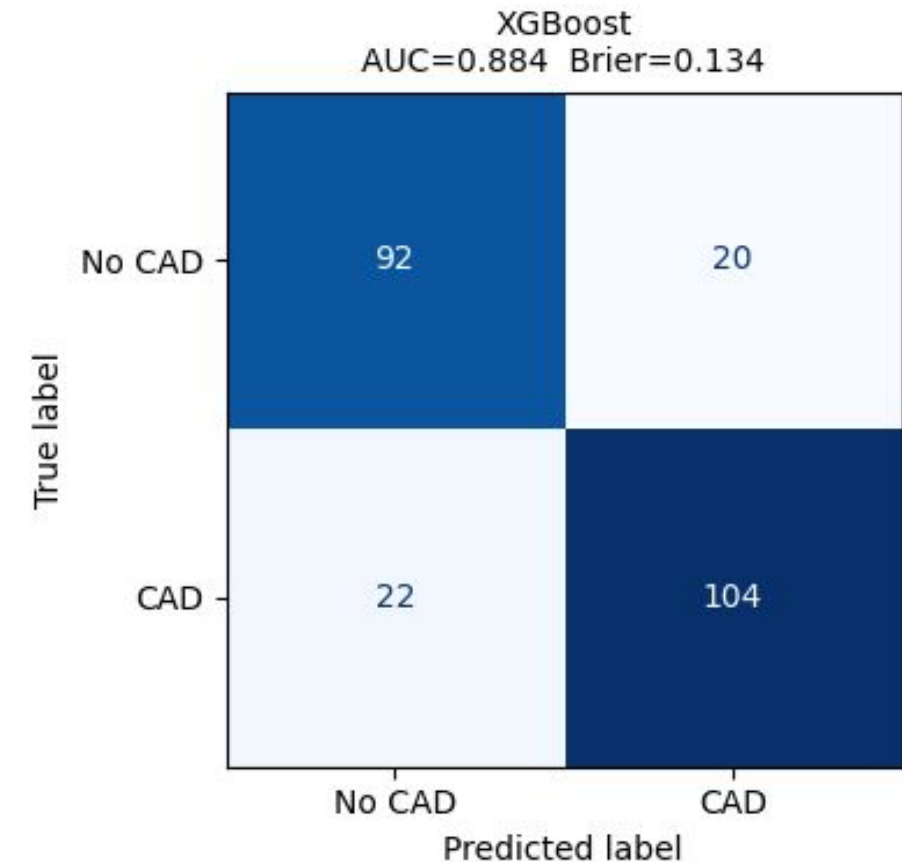
Results : Clinical Dataset

- XGBoost achieves the highest AUC (approximately 0.88), indicating the best discrimination performance
- All models perform significantly better than a random baseline (strong overall predictive ability)
- ROC curves show consistent performance across different thresholds



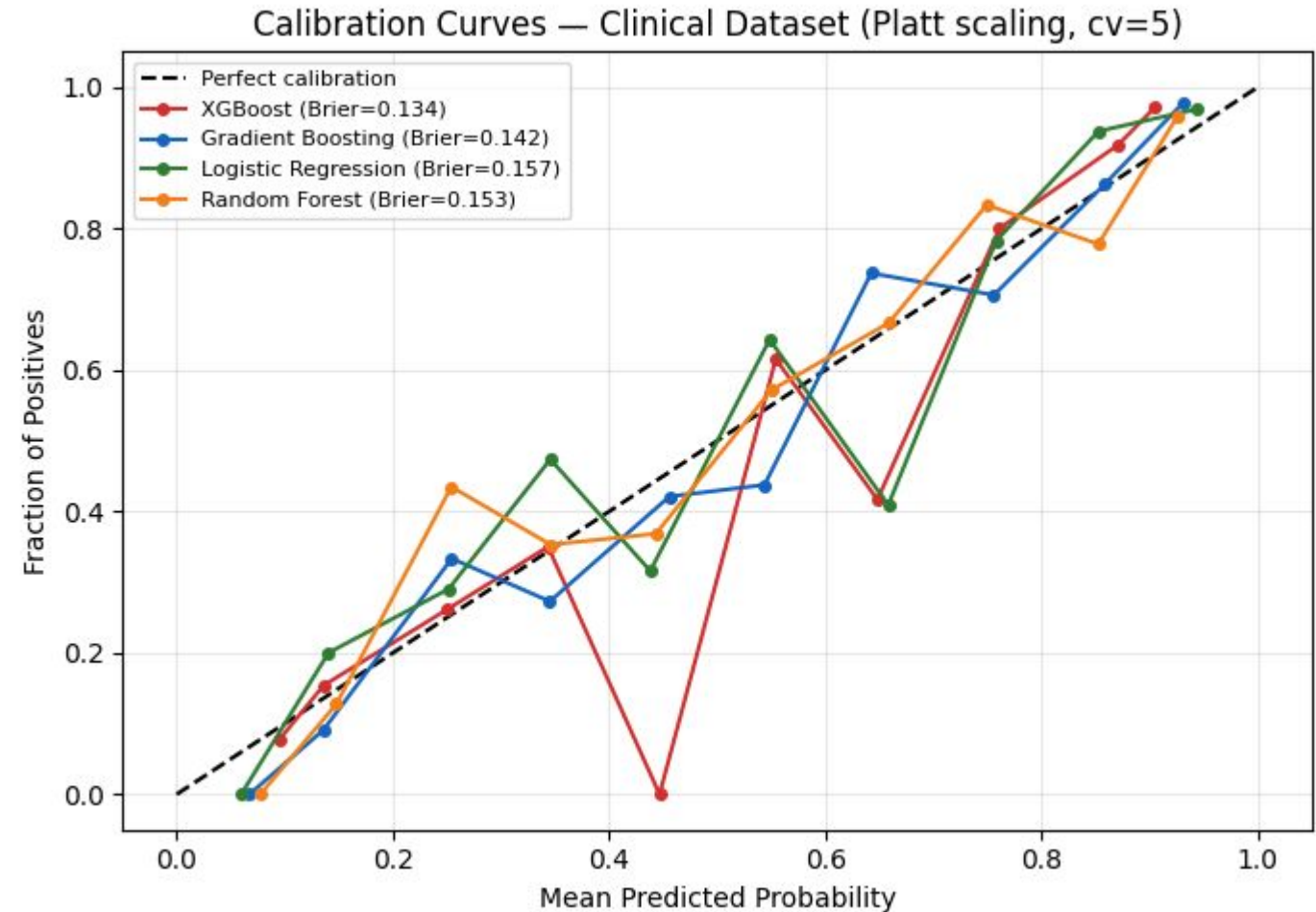
Results: Confusion Matrix Analysis

- Correct predictions: 92 (No CAD) and 104 (CAD) implying good overall classification
- False positives: 20 (predict CAD when none), there are some over-diagnosis
- False negatives: 22 (missed CAD cases). This is critical but relatively low.
- Balanced performance, model maintains good precision and recall.



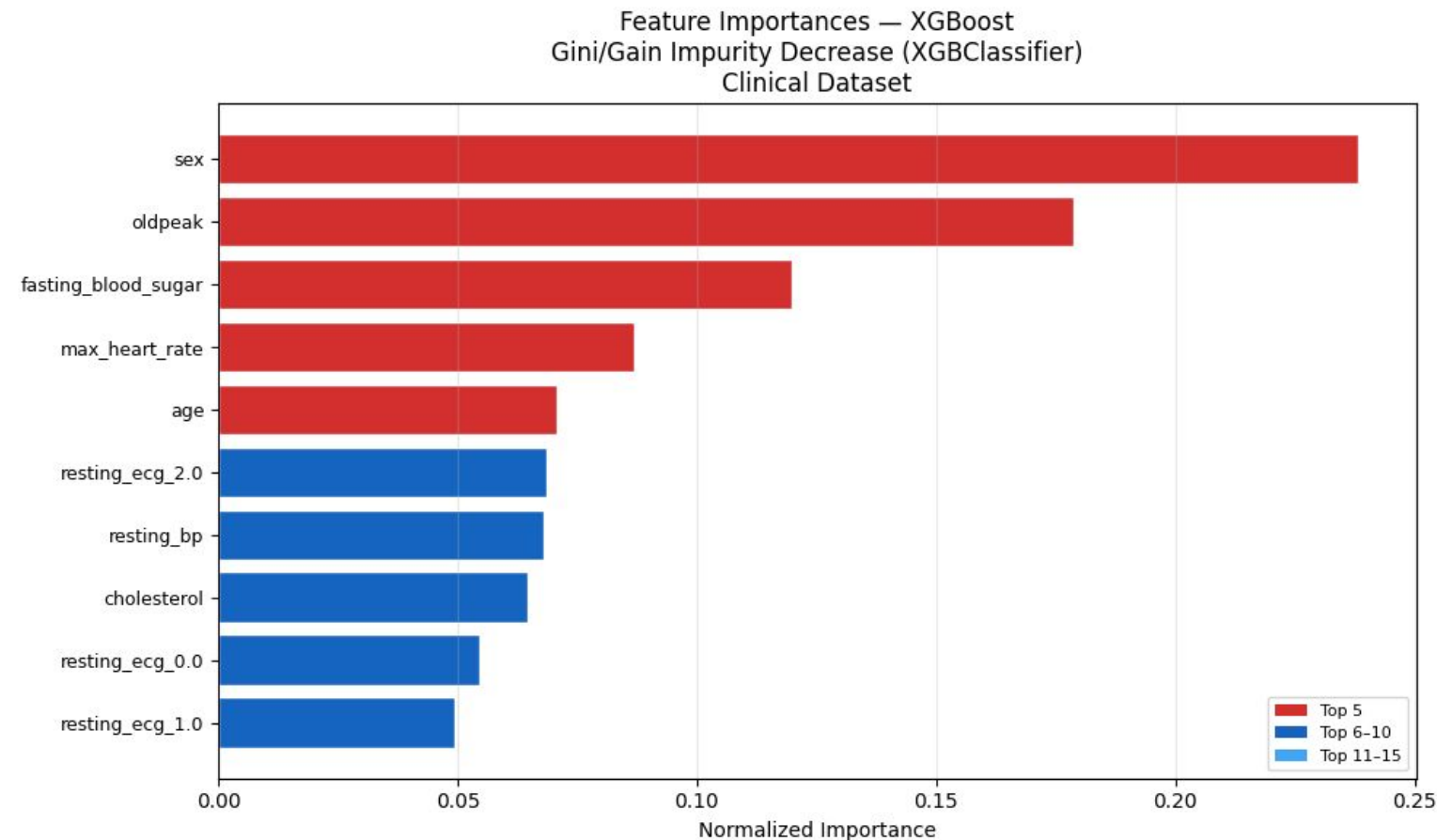
Results : Clinical Dataset

- XGBoost achieves the lowest Brier score (around 0.13), indicating the best calibration
- Predicted probabilities closely align with the ideal diagonal, showing high reliability
- Other models show slight deviations, suggesting less accurate probability estimates



Results : Clinical Model - Feature Importance Analysis

- Top predictors include sex, oldpeak, and fasting blood sugar
- Clinical features like heart rate, BP, cholesterol, and ECG are well captured
- Importance is distributed across features, not dominated by a single variable
- Findings align with known medical risk factors for CAD

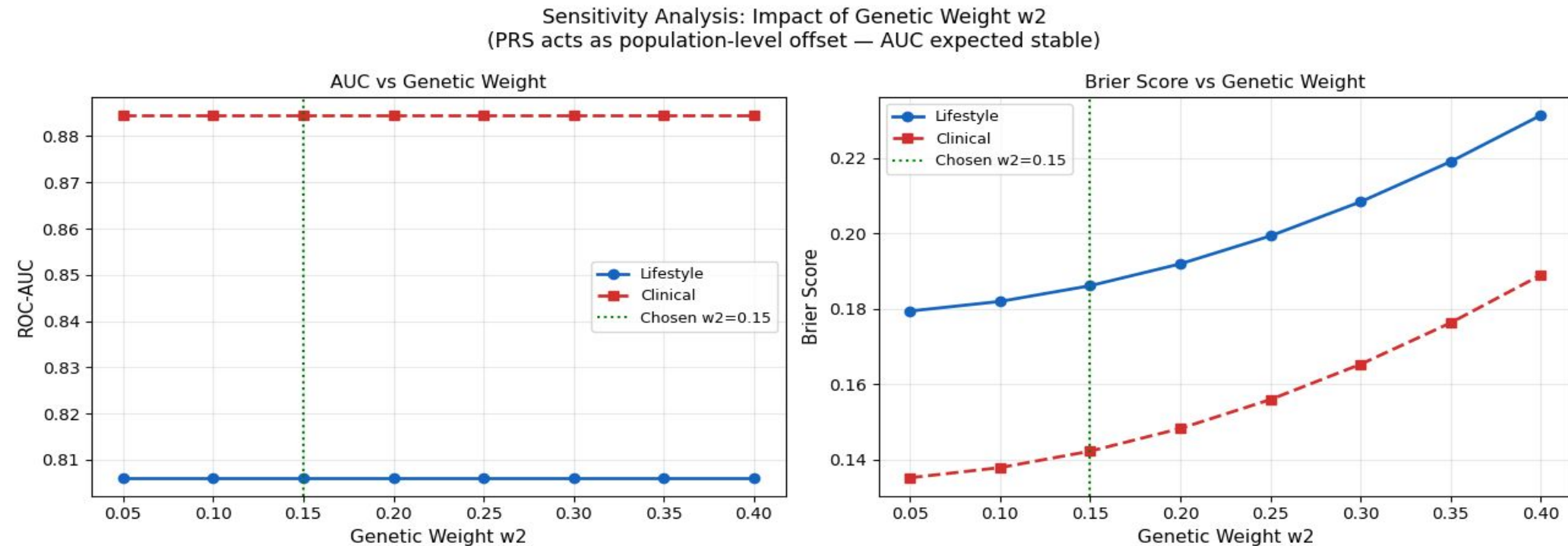


Results : Impact of PRS Integration

Base Models vs PRS-Integrated Predictions				
Component	AUC	Brier	Mean Prob	Range
Lifestyle	0.81	0.178	0.49	0.09–0.90
Clinical	0.88	0.134	0.52	0.09–0.91
Lifestyle + PRS	—	—	0.57	0.23–0.92
Clinical + PRS	—	—	0.59	0.23–0.92

- PRS acts as a population-level calibration factor, not a discriminative feature
- Integration results in an upward shift in predicted risk (approximately +0.07 mean probability)
- AUC and ranking performance remain unchanged, confirming no impact on discrimination
- Weighted fusion (85% model, 15% PRS) enables calibrated and population-aware risk estimates

Results : Sensitivity Analysis



- AUC remains constant across all w_2 values, indicating no impact on discrimination
- Brier score increases as w_2 increases, showing reduced calibration with higher genetic weight
- Optimal balance observed at $w_2 \approx 0.15$, maintaining performance with minimal calibration loss

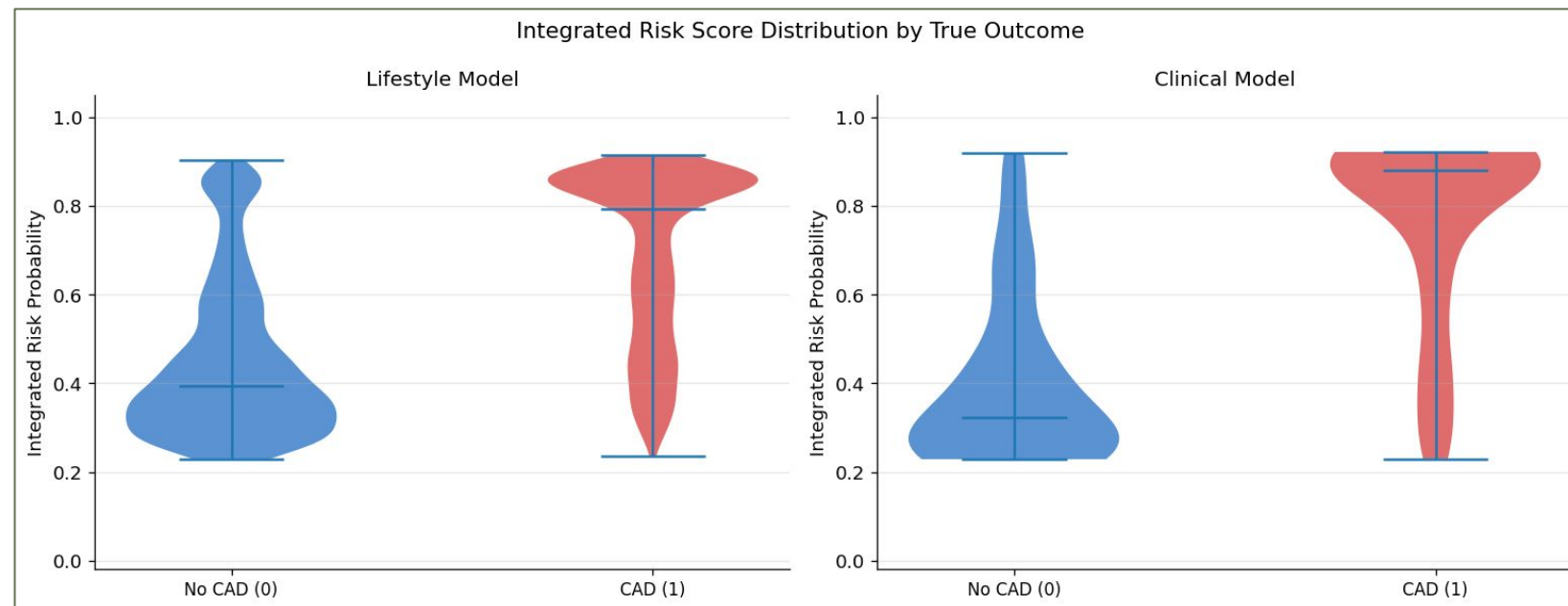
Results: Risk Stratification and PRS Impact

Risk Stratification & PRS Impact		
Metric	Lifestyle	Clinical
Reclassified by PRS	32.1%	25.2%
AUC Change	0.00	0.00
Brier Change	+0.0076	+0.0081

- PRS leads to significant patient reclassification (approximately 25–32%)
- AUC remains unchanged, indicating preserved discrimination
- Slight increase in Brier score reflects a calibration trade-off
- Integration enables more personalized and population-aware risk estimates

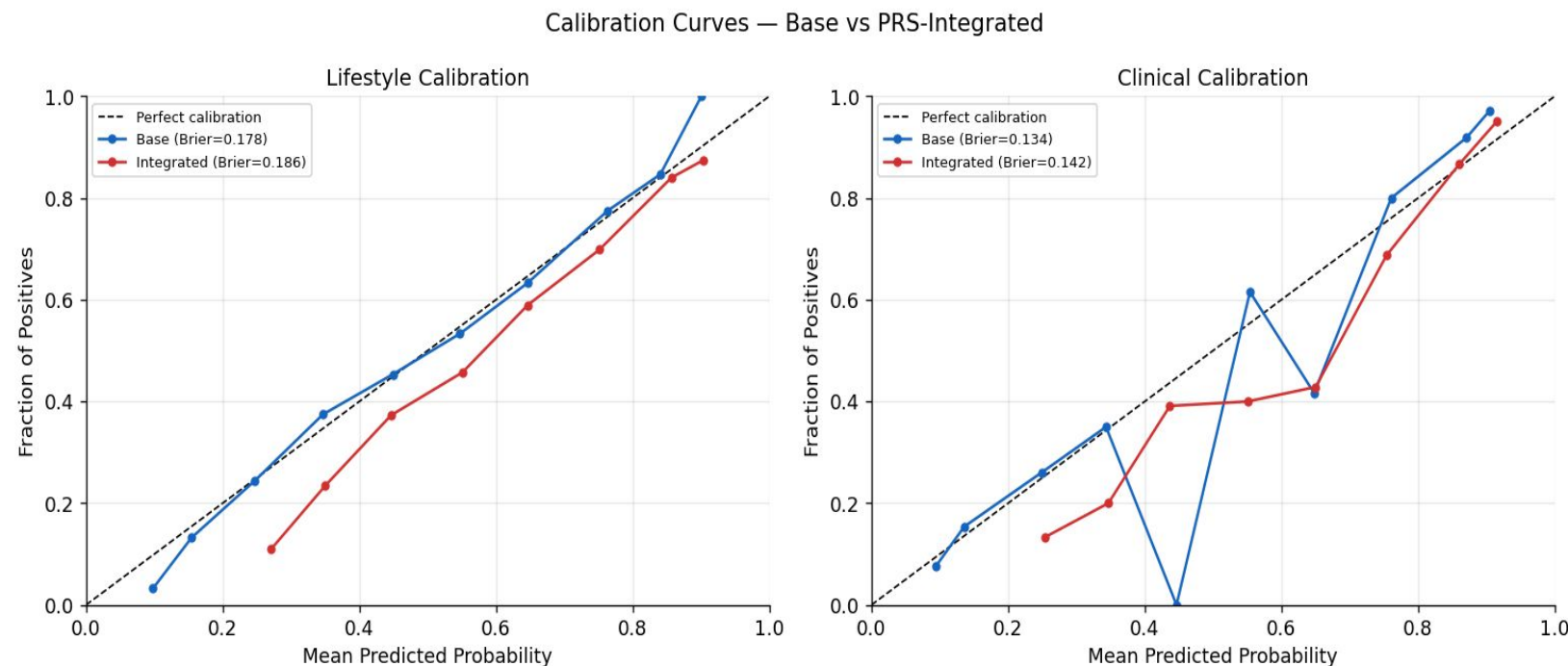
Results : Risk Score Distribution by Outcome

- Clear separation between No CAD (low risk) and CAD (high risk) groups
- CAD cases are concentrated in the higher probability range (around 0.8–0.9)
- Minimal overlap between distributions indicates strong discrimination
- Clinical model shows slightly sharper separation compared to the lifestyle model



Results: Calibration Impact of PRS Integration

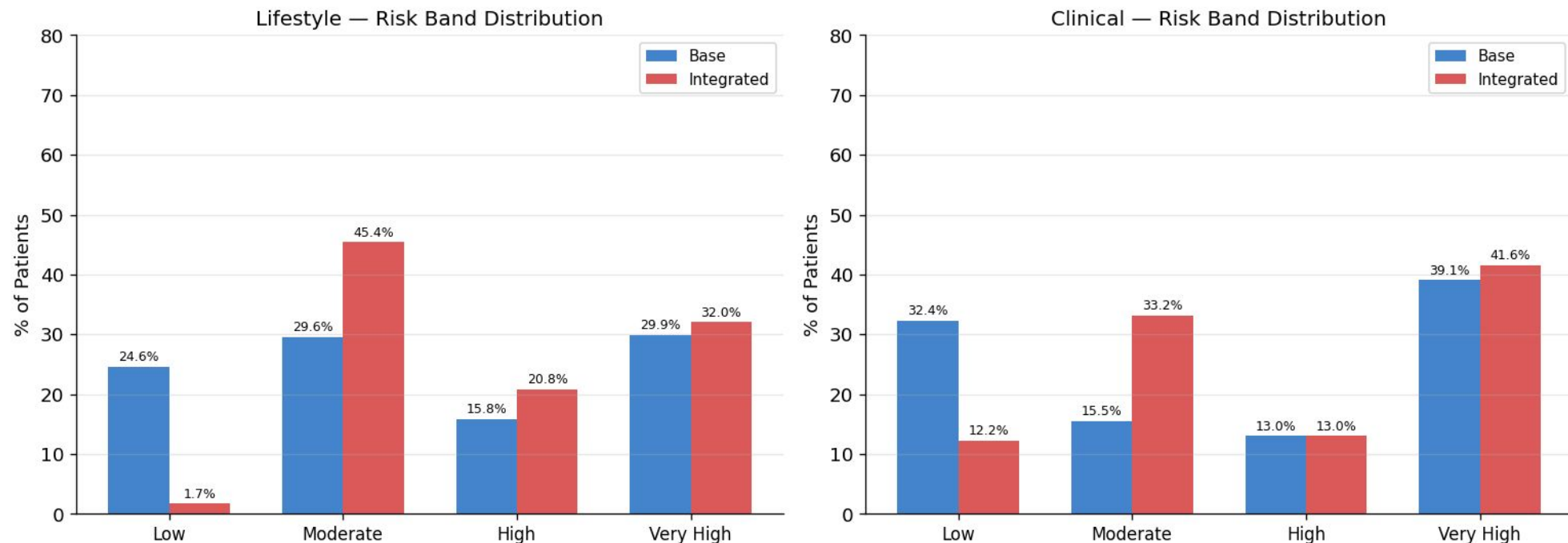
- AUC remains unchanged, indicating no effect on model discrimination
- Slight increase in Brier score reflects a minor calibration trade-off
- Calibration curves remain close to the ideal diagonal, showing reliable predictions
- PRS acts as a probability shift (calibration offset), not a predictive feature



Results : Impact of PRS on Risk Stratification

- PRS integration shifts patients from low to moderate and high risk categories
- Increase observed in moderate and very high risk groups after integration
- Lifestyle model shows a larger shift (around 32% reclassification) compared to clinical (around 25%)
- Enables more refined and personalized risk stratification

Risk Stratification: Band Distribution Before & After PRS Integration

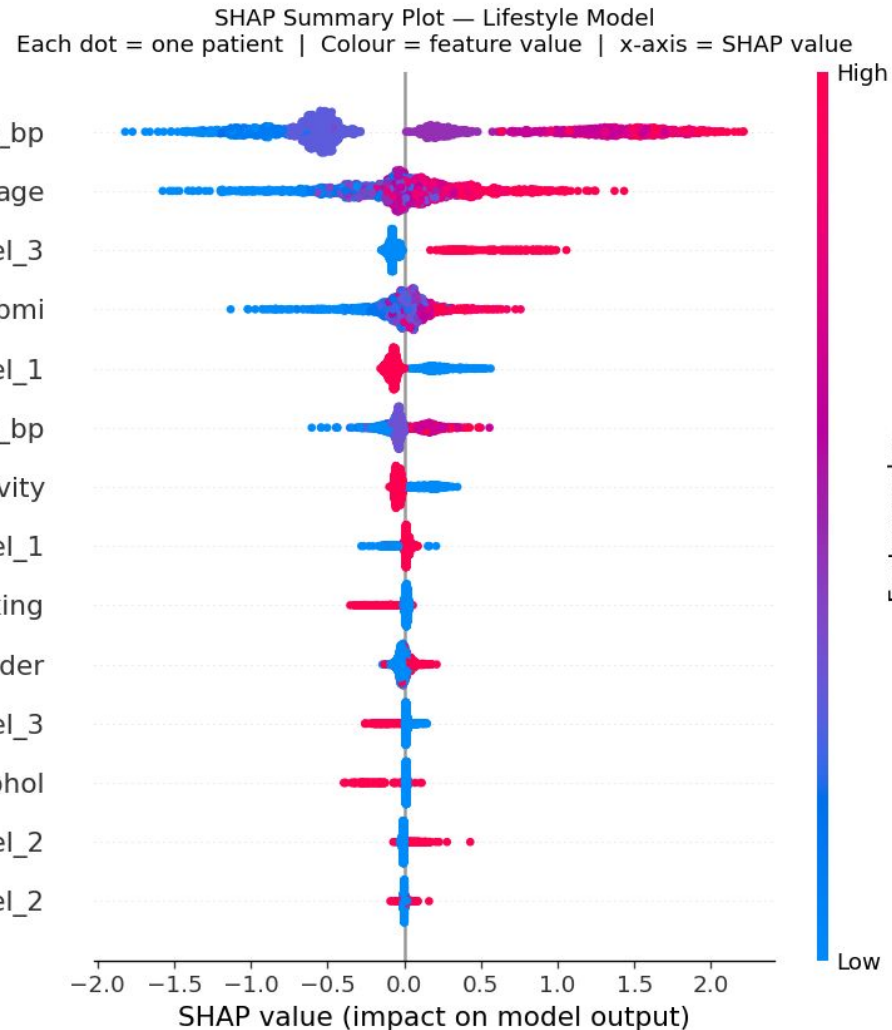


Results: Final Calibration and Model Refinement

Final Calibration & Model Refinement			
Cohort	Model	AUC	Brier
Lifestyle	Base	0.806	0.178
	Integrated	0.806	0.186
	Calibrated	0.806	0.179
Clinical	Base	0.885	0.134
	Integrated	0.885	0.142
	Calibrated	0.885	0.136

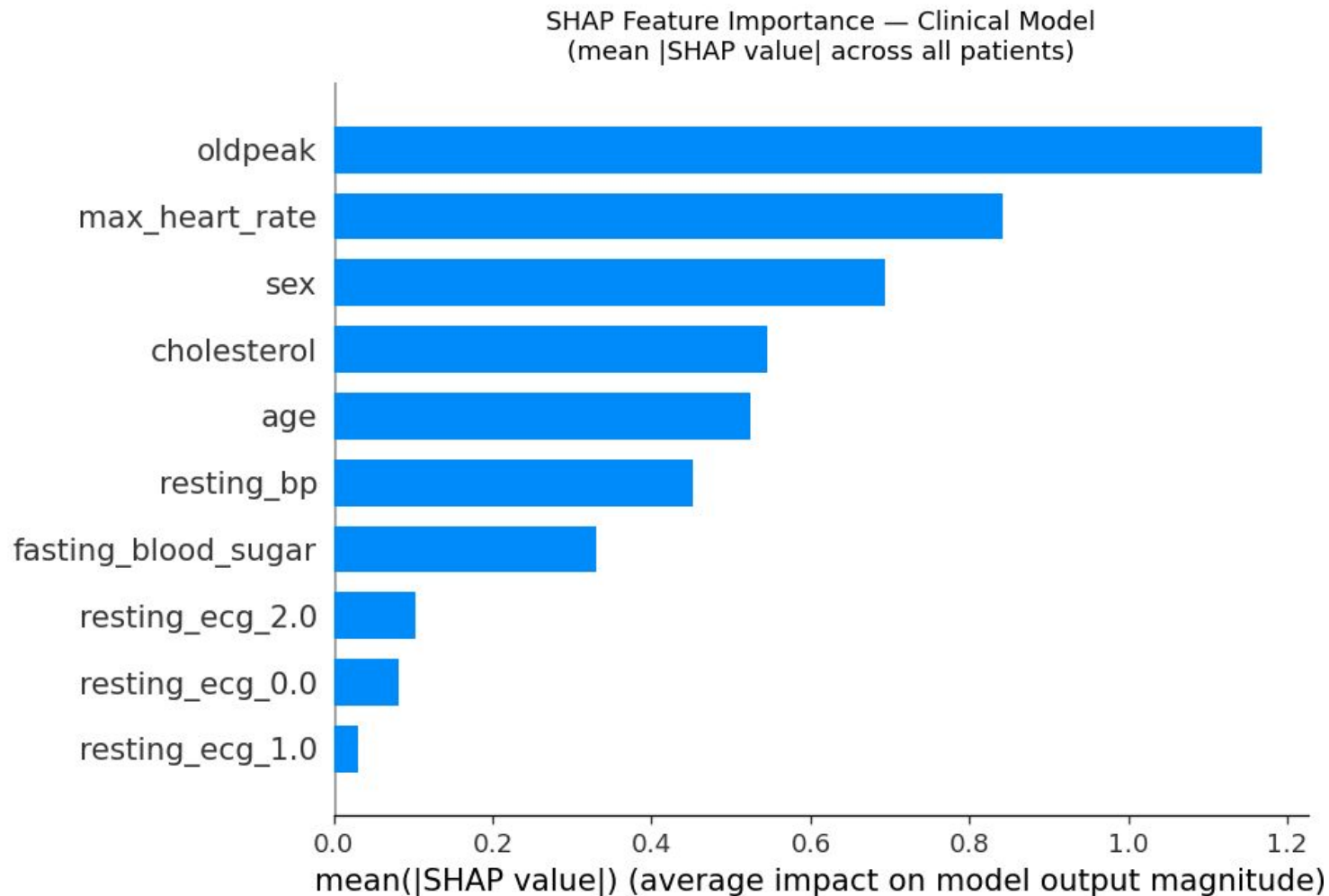
- Calibration improves Brier score, leading to more reliable probability estimates
- AUC remains unchanged, indicating preserved model discrimination
- PRS integration initially increases Brier score due to probability shift
- Platt scaling corrects this shift and restores prediction confidence

Results: SHAP Analysis – Lifestyle Model



- Systolic blood pressure and age are the strongest drivers of lifestyle-based risk
- Higher feature values (red) increase predicted CAD risk (positive SHAP contribution)
- Lower feature values (blue) decrease predicted risk (negative SHAP contribution)
- Features show consistent directional impact, indicating stable model behavior

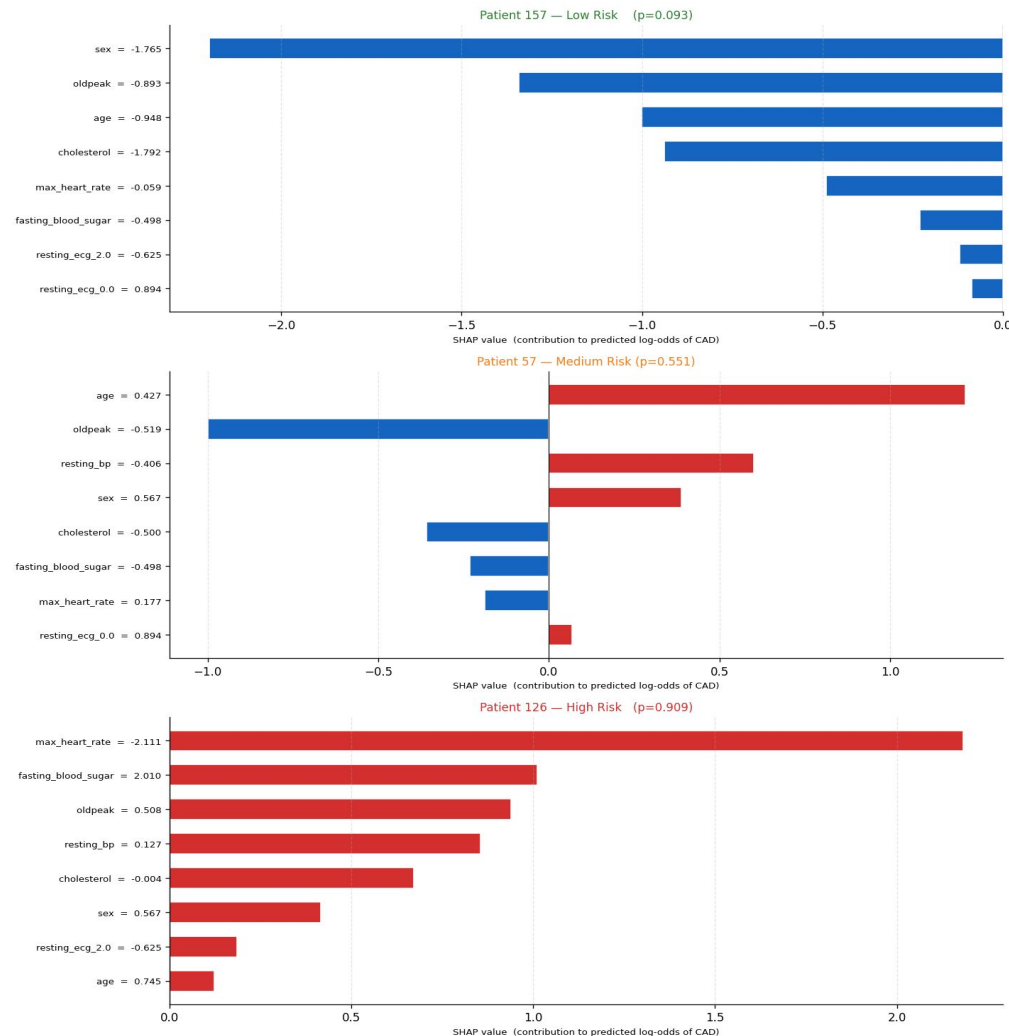
Results: SHAP Analysis – Clinical Model



- Oldpeak is the most influential feature, followed by maximum heart rate and sex
- Model captures key cardiac indicators including ECG changes, heart rate, blood pressure, and cholesterol
- Feature importance is distributed across multiple clinical variables, indicating a multi-factor model
- Lower-ranked ECG features contribute minimally to predictions

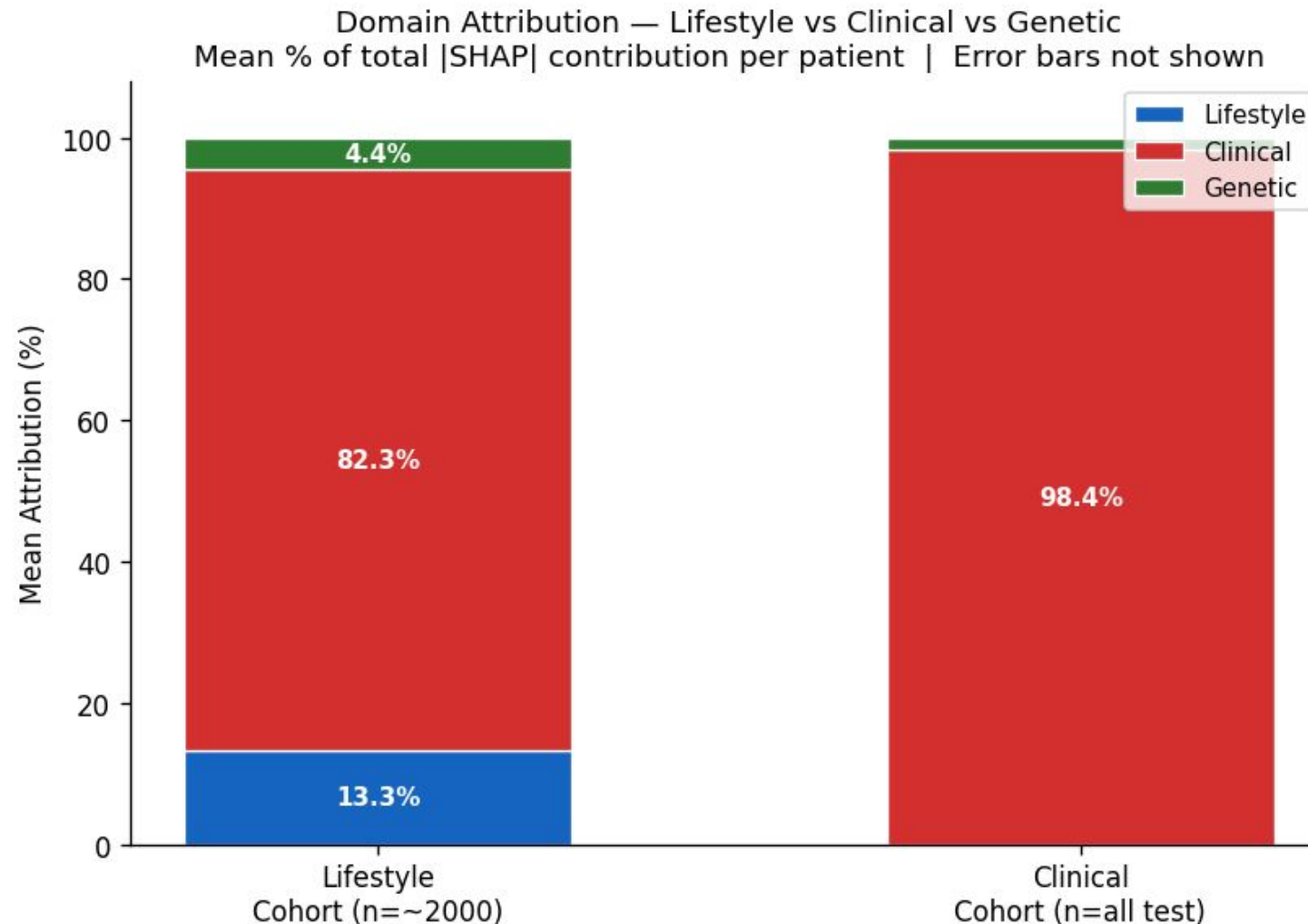
Results: SHAP Explanation Across Risk Levels

SHAP Waterfall Plots — Clinical Model
Top-8 Feature Contributions (Low / Medium / High Risk)



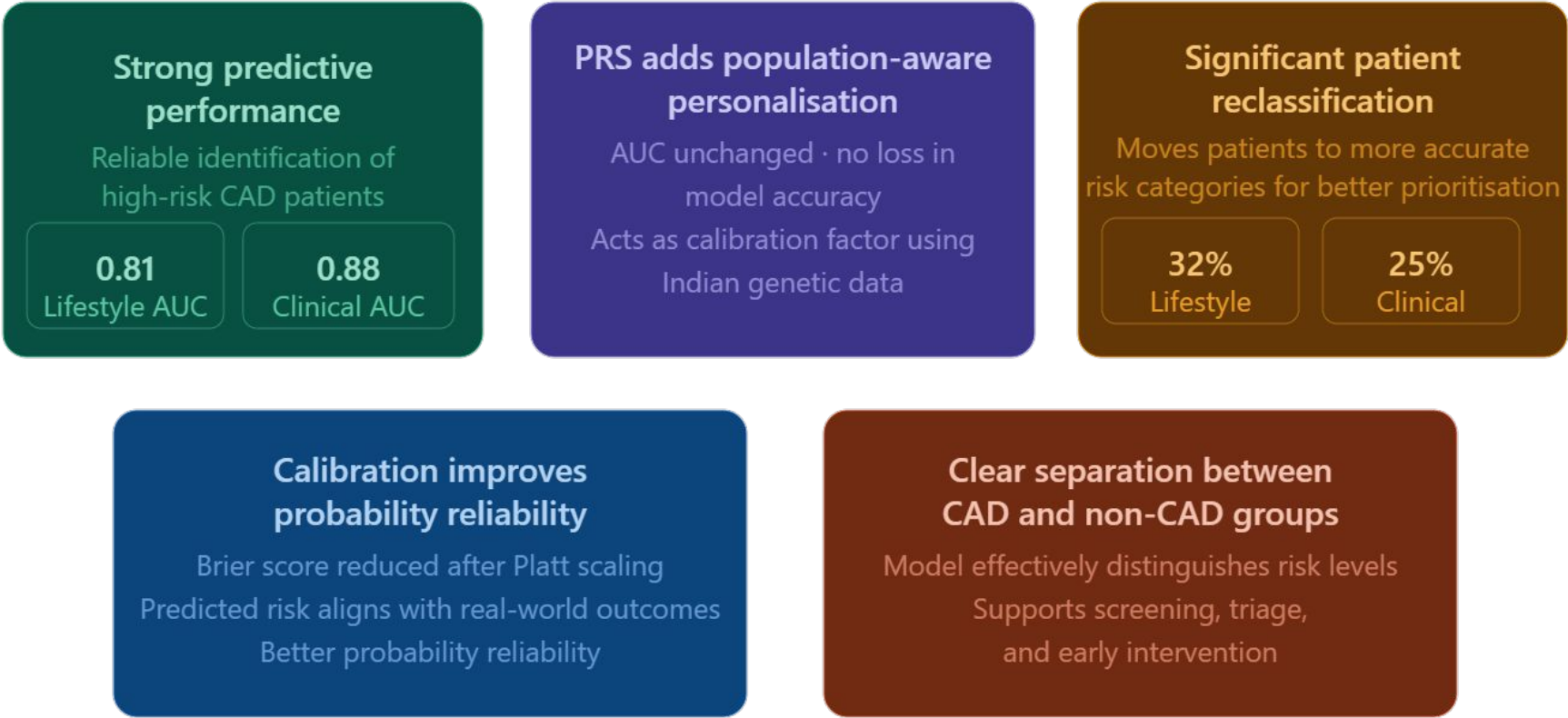
- Feature contributions show how individual variables increase (red) or decrease (blue) predicted risk
- Low-risk patient: most features contribute negatively, resulting in lower predicted risk
- High-risk patient: key features such as heart rate, blood sugar, and oldpeak strongly increase risk
- Medium-risk patient: mixed feature contributions lead to a balanced prediction

Results: Domain-Level Contribution Analysis



- Clinical features dominate predictions, contributing approximately 82–98%
- Lifestyle factors provide moderate additional influence, around 13%
- Genetic (PRS) contribution is small, approximately 1–4%, acting as a modifier
- Confirms system design: clinical core, supported by lifestyle, with genetic adjustment

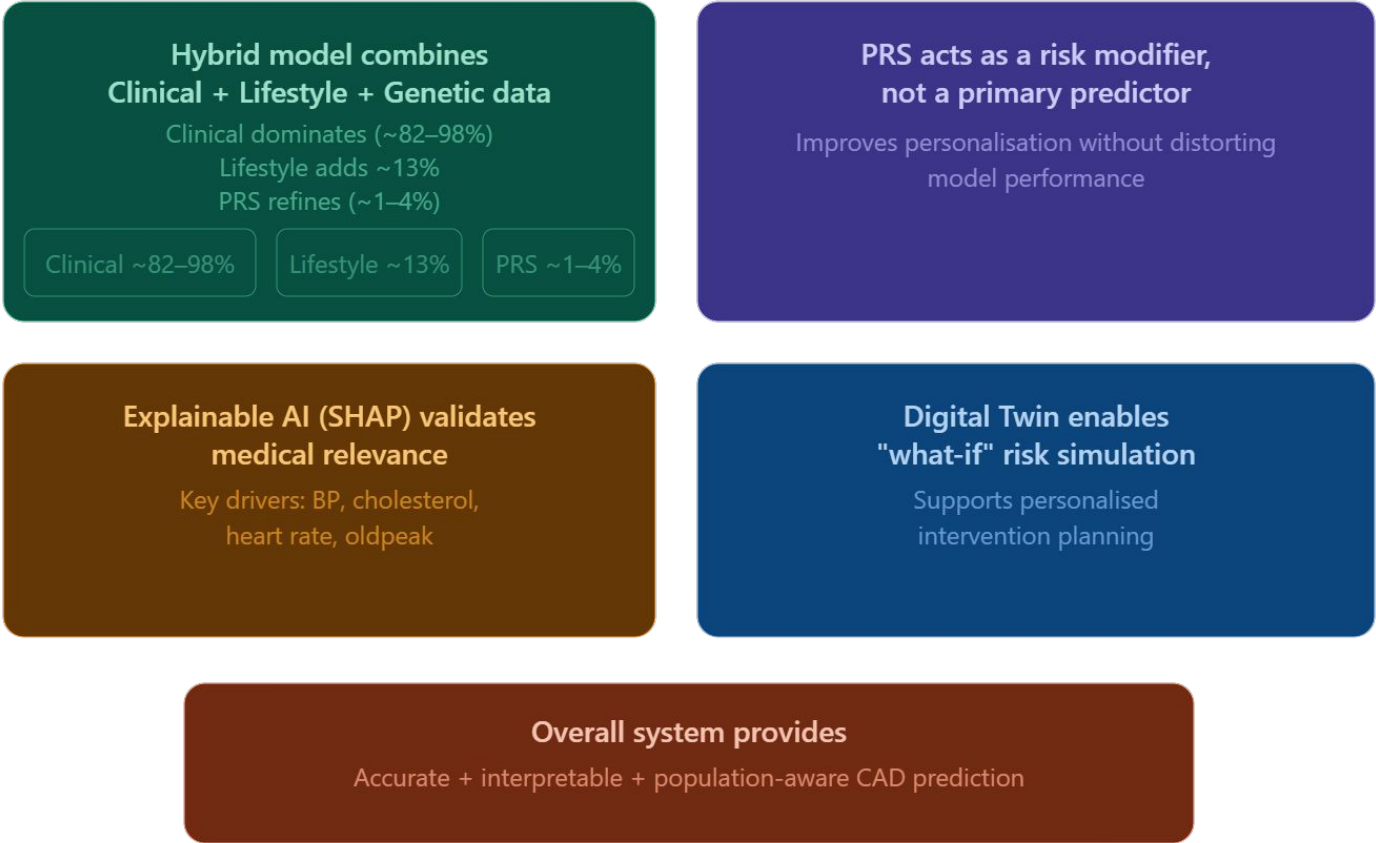
Conclusion: Final Results & Practical Impact



CAD risk prediction pipeline · key findings

- Model performance
- PRS integration
- Reclassification
- Calibration
- Risk separation

System Insights and Contribution



CAD hybrid model · architecture & capabilities

■ Hybrid model ■ PRS role ■ Explainability ■ Digital Twin ■ Overall system

Individual contribution

Karan Bhojwani

- Contributed to data preprocessing and dataset preparation
- Worked on data cleaning and feature setup for both pipelines

Saanvi Kakkar

- Handled major system development including PRS computation and integration
- Worked on model explainability, calibration, and results analysis
- Contributed to core implementation and report writing

Sanika Gupta

- Handled major system development including model building and experimentation
- Contributed to preprocessing, validation, and results interpretation
- Assisted in report writing and documentation

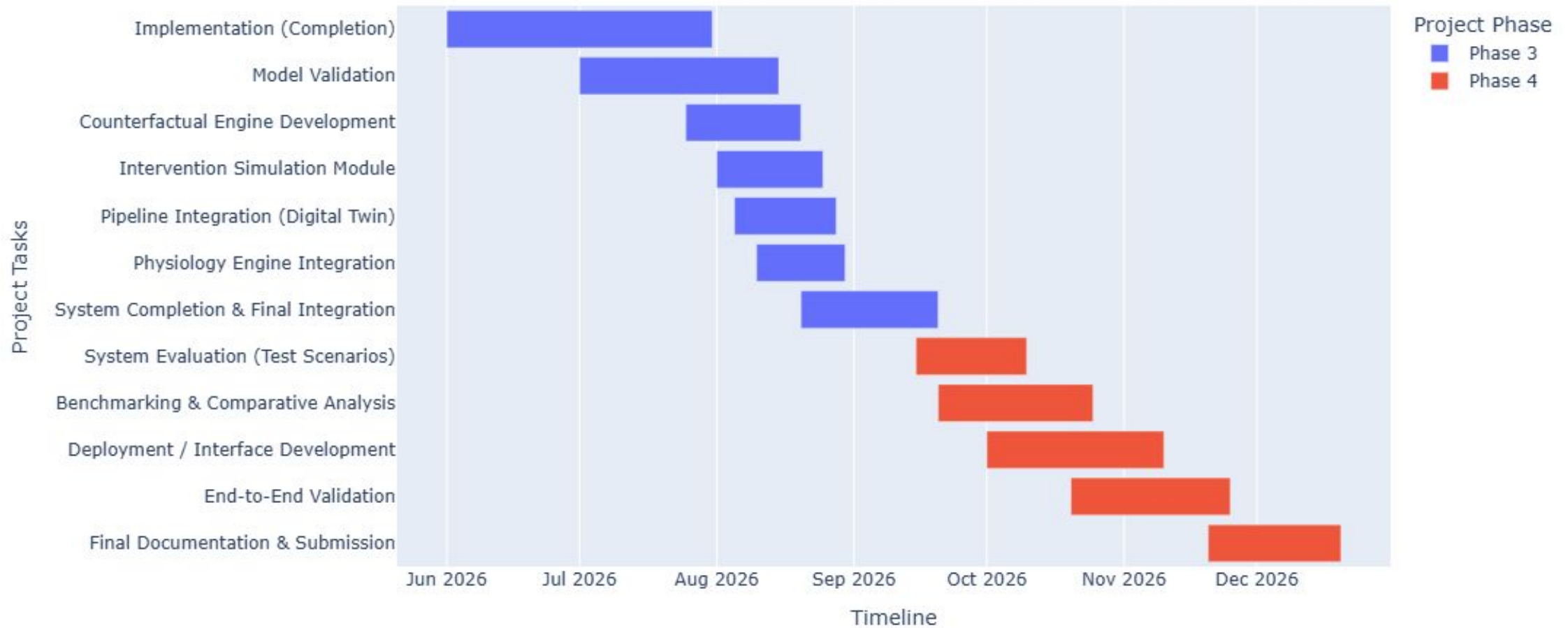
Shrest Kumar

- Contributed to data preprocessing and initial pipeline setup
- Assisted in data preparation and documentation

References

- [1] A. V. Khera, M. Chaffin, K. G. Aragam, M. E. Haas, C. Roselli, S. H. Choi, et al., “Genome-wide polygenic scores for common diseases identify individuals with risk equivalent to monogenic mutations,” *Nature Reviews Cardiology*, vol. 19, no. 4, pp. 225–238, 2022.
- [2] A. G. Hindy, P. Natarajan, and S. Kathiresan, “Ancestry-specific polygenic risk scores and improvement of coronary artery disease prediction,” *Nature Communications*, vol. 14, Art. no. 1023, 2023.
- [3] GenomeIndia Consortium, “The GenomeIndia Project: Cataloguing genetic diversity of the Indian population,” *Nature Genetics*, vol. 57, pp. 1–10, 2025.
- [4] S. Corral-Acero, F. Margara, E. Marciniak, et al., “The ‘Digital Twin’ to enable the vision of precision cardiology,” *npj Digital Medicine*, vol. 3, Art. no. 1, pp. 1–10, 2022.

Project Progress Plan for Phase 3 and Phase 4



Thank You